Sathyabaarathi Shanthi Ravichandran
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PROFILE SUMMARY: Systems immunologist focused on how aging reshapes immune responses to infection and vaccination. Skilled in multi-omic data integration, network biology, and translational biomarker discovery to improve vaccine efficacy and immune health in older adults.

POSITIONS

- **Postdoctoral Associate (The Ucar Lab)**
  The Jackson Laboratory for Genomic Medicine, Farmington, CT, USA
  (Oct 2021 – Present)
- **Research Associate (Chandra Lab)**
  Indian Institute of Science, Bangalore, India
  (Sep 2020 – Sep 2021)
- **Visiting Student**
  University of Leeds, Department of Applied Mathematics, Leeds, UK
  (Oct 2016 – Dec 2016)
- **Ph.D. Research Scholar**
  Indian Institute of Science, Bangalore, India
  (Aug 2013 – Aug 2020)

EDUCATION

- **Ph.D. in Science**
  Indian Institute of Science, Bangalore, India (Aug 2013 – Aug 2020)

- **B.Tech. in Bioinformatics**
  SASTRA University, Thanjavur, India (Jun 2006 – May 2010)

AWARDS & HONORS

- **Robert E. Leet & Clara Guthrie Patterson Trust Mentored Research Award** (2026–2028), Health Resources in Action, Boston, MA
- **Brooks Scholar Postdoctoral Fellowship** (2023–2024), The Jackson Laboratory Cancer Center
- **AAI Trainee Abstract Award (2023),** American Association of Immunologists – IMMUNOLOGY 2023
- **Best Poster Award**, 32nd Annual Short Course on Experimental Models of Human Cancer (2023)
- **Best Poster Award**, Sex Differences in the Immune System Meeting (2022)

- **Global Health Travel Award**, Keystone Symposia – Translational Systems Immunology (2018)
- Global Travel Award, Boehringer Ingelheim Fonds (2016)
- Dean's List for Academic Excellence, SASTRA University (2008–2010)

PEER-REVIEWED PUBLICATIONS

https://scholar.google.com/citations?user=uhoSPP4AAAAJ&hl=en

1. **Ravichandran S**, Erra-Diaz F, Karakaslar OE, Marches R, Kenyon-Pesce L, Rossi R, Chaussabel D, Nehar-Belaid D, LaFon DC, Pascual V, Palucka K, Ucar D, Banchereau J, Kuchel GA. Distinct baseline immune characteristics associated with responses to conjugated and unconjugated pneumococcal polysaccharide vaccines in older adults. ***Nature Immunology***. 2024; 25(1):1–4. DOI: 10.1038/s41590-023-01717-5 **(Impact Factor: 27.6)**
2. **Ravichandran S**, Banerjee U, Devi G, Kandukuru R, Thakur C, Chakravortty D, Narayanaswamy B, Singh A, Chandra N. VB10, a new blood biomarker for differential diagnosis and recovery monitoring of acute viral and bacterial infections. ***EBioMedicine***. 2021; 67:103352. DOI: 10.1016/j.ebiom.2021.103352 **(Impact Factor: 10.8)**
3. Nehar-Belaid D*, Sokolowski M*, **Ravichandran S***, Banchereau J, Chaussabel D, Ucar D. Baseline immune states (BIS) associated with vaccine responsiveness and factors that shape the BIS. ***Seminars in Immunology***. 2022; 70:101842. **(Co-first author)** DOI: 10.1016/j.smim.2023.101842 **(Impact Factor: 7.8)**
4. Thakur C, Tripathi A, **Ravichandran S**, Shivananjaiah A, Chakraborty A, Varadappa S, Chikkavenkatappa N, Nagarajan D, Lakshminarasimhaiah S, Singh A, Chandra N. A new blood-based RNA signature (R_9), for monitoring effectiveness of tuberculosis treatment in a South Indian longitudinal cohort. **iScience**. 2022 Jan 10;25(2):103745. DOI: 10.1016/j.isci.2022.103745 **(Impact Factor: 4.1)**
5. Nagarajan D, Roy N, Kulkarni O, Nanajkar N, Datey A, **Ravichandran S**, Thakur C, Aprameya IV, Sarma SP, Chakravortty D. Ω76: A designed antimicrobial peptide to combat carbapenem- and tigecycline-resistant *Acinetobacter baumannii*. ***Science Advances***. 2019; 5(7):eaax1946. DOI: 10.1126/sciadv.aax1946 **(Impact Factor: 12.5)**
6. **Ravichandran S**, Chandra N. Interrogation of genome-wide networks in biology: comparison of knowledge-based and statistical methods. ***Int. J. Advances in Engineering Sciences and Applied Mathematics***. 2019; 11(2):119–137. https://doi.org/10.1007/s12572-018-0242-9 **(Impact Factor: 0.8)**
7. Nagarajan D, Nagarajan T, Roy N, Kulkarni O, **Ravichandran S**, Mishra M, Chakravortty D, Chandra N. Computational antimicrobial peptide design and evaluation against multidrug-resistant clinical isolates of bacteria. ***Journal of Biological Chemistry***. 2018; 293(10):3492–3509. DOI: 10.1074/jbc.M117.805499 **(Impact Factor: 3.9)**
8. Sreenivasan, S., **Ravichandran, S.**, Vetrivel, U., & Krishnakumar, S. (2013). Modulation of multidrug resistance 1 expression and function in retinoblastoma cells by curcumin. ***Journal of Pharmacology and Pharmacotherapeutics***, *4*(2), 103-109. DOI: 10.4103/0976-500X.110882 **(Impact Factor: 0.8)**
9. Vetrivel U, **Ravichandran S**, Kuppan K, Mohanlal J, Das UN, Narayanasamy A. Agonistic effect of polyunsaturated fatty acids (PUFAs) on brain-derived neurotrophic

factor (BDNF). ***Lipids in Health and Disease***. 2012; 11(1):1–8. DOI: 10.1186/1476-511X-11-109 **(Impact factor: 4.2)**

10. Harishankar A, Jambulingam M, Gowrishankar R, Vetrivel U, **Ravichandran S**, et al. Phylogenetic comparison of HSV1 isolates: exonic US4, US7, UL44 regions. ***Virology Journal***. 2012; 9(1):1–8. DOI: 10.1186/1743-422X-9-65. **(Impact Factor: 4.0)**
11. Vinita K, Sripriya S, Prathiba K, Vaitheeswaran K, **Ravichandran S**, et al. ICAM-1 K469E polymorphism and diabetic retinopathy. *BMJ Open*. 2012; 2(4):e001036. DOI: 10.1136/bmjopen-2012-001036 **(Impact Factor: 2.4)**
12. Sreenivasan S, **Ravichandran S**, Umashankar V, Subramanian K. Inhibitory effects of curcumin on MRP1 in retinoblastoma cells. ***Bioinformation***. 2012; 8(1):13. DOI: 10.6026/97320630008013 **(Impact Factor: 1.9)**
13. Jambulingam M, Vetrivel U, **Ravichandran S**, et al. Functional characterization of UL54 mutations in HCMV. ***Bioinformation***. 2011; 5(9):390. doi: 10.6026/97320630005390 **(Impact Factor: 1.9)**
14. Gandra M, Umashankar V, Kasinathan N, Krishnan T, **Ravichandran S**, et al. Molecular screening of CYP4V2 gene in Bietti crystalline dystrophy. ***Molecular Vision***. 2011; 17:1970. PMID: 21850171 **(Impact Factor: 1.4)**

CONFERENCE PROCEEDINGS (Journal Supplements)

15. **Ravichandran S**, Ucar D, Marches R, Chaussabel D, Kuchel GA, Banchereau J. Variation in the single-cell epigenomic and transcriptomic profiles in response to seasonal influenza vaccination: a comparison between healthy young and older adults. *The Journal of Immunology*. 2024; 212(1_Suppl):1566–5781. **(Impact Factor: 3.3)**
16. **Ravichandran S**, Erra-Diaz F, Karakaslar OE, Marches R, Rossi R, Nahm MH, Chaussabel D, Kuchel GA, Banchereau J, Ucar D. Pre-vaccination CD56dim CD16+ NK cell abundance and Th1/Th17 ratio predict responsiveness to conjugated pneumococcal vaccine in older adults. *The Journal of Immunology*. 2023; 210(Suppl_1):252.03. **(Impact Factor: 3.3)**
17. Ucar D, Kuchel GA, Banchereau J, **Ravichandran S**, Karakaslar OE, Erra-Diaz F, Marquez E, Marches R. Sexual Dimorphism in Human Immune System Aging. *Innovation in Aging*. 2022; 6(Suppl_1):165. **(Impact Factor: 4.3)**

PREPRINTS

18. Grabauskas T, Trinity L, Verschoor CP, Marches R, Thibodeau A, Nehar-Belaid D, Eryilmaz G, Mahajan AS**, Ravichandran S**, et al. CMV-specific clonal expansion of Th1, GZMK$^+$ CD8$^+$, and TEMRA T cells revealed by human PBMC single-cell profiling. Under review (Genome Biology), **2025**. **(Impact Factor: 9.1)**
19. **Ravichandran S**, Bhatt B, Shah A, Das D, Balaji KN, Chandra N. Knock-down of a regulatory barcode shifts macrophage polarization from M1 to M2 and increases pathogen burden upon S. aureus infection. ***eLife*** **(reviewed preprint).**

PATENTS

1. **Ravichandran S**, Chandra N. Biomarkers and Associated Methods for Detecting Bacterial and Viral Infections in a Blood Sample. Indian Patent No. 404211, filed Apr 10 2020, granted Aug 23 2022. (Granted)
2. **Ravichandran S**, Ucar D, Banchereau J. Methods for Assessing Pneumococcal Vaccine Responsiveness. U.S. Provisional Patent Application No. 63/741,702, filed Jan 3 2025. (Pending)

INVITED TALKS & PRESENTATIONS

1. Genomic Signatures of Acute Infections and Vaccine Responsiveness. RAMAIAH UNIVERSITY OF APPLIED SCIENCES, Bangalore, India. 2nd December, 2025. **(Invited Talk)**

2. Distinct Baseline Immune Characteristics Associated with Responses to Conjugated and Unconjugated Pneumococcal Polysaccharide Vaccines in Older Adults. ***8th Northeast Glenn Symposium on the Biology of Aging***, **Yale University,** Oct 2024 (**Podium Presentation**).
3. A single-cell transcriptome atlas of Fluzone responses in healthy young and older adults. ***Systems Immunology in Aging and Complex Diseases***, The Jackson Laboratory, Farmington, CT, June 2024 (**Invited Talk; Co-Organizer**).
4. Distinct Immune Characteristics Associated with Vaccine Responsiveness in Older Adults. ***AAI Annual Meeting – IMMUNOLOGY 2023***, **Washington, DC (Podium & Poster; AAI Trainee Abstract Award 2023).**
5. Identification of a Robust Blood-Based Biomarker Signature for Chronic Systemic Inflammation. ***Keystone Symposia – Translational Systems Immunology***, Snowbird, UT, 2018 **(Podium & Poster; Global Travel Award).**

POSTER PRESENTATIONS (Selected)

1. Activation of cDC2 and monoDCs on Day 1 is associated with stronger responses to Fluzone HD in older adults. ***9th Northeast Glenn Foundation Symposium***, **University of Connecticut, 2025.**
2. Distinct baseline immune characteristics associated with responses to conjugated and unconjugated pneumococcal vaccines in older adults. ***American Geriatrics Society (AGS) Annual Meeting***, **2024 (Presidential Poster Session).**
3. Pre-vaccination CD56dim CD16+ NK cells and Th1/Th17 ratio predict responsiveness to conjugated pneumococcal vaccine in older individuals. ***32nd Annual Short Course on Experimental Models of Human Cancer***, **The Jackson Laboratory, Bar Harbor, ME, 2023 (Best Poster Award).**
4. Pre-vaccination CD56dim CD16+ NK cells and Th1/Th17 ratio predict responsiveness to conjugated pneumococcal vaccine in older individuals. ***Sex Differences in the Immune System***, **2022 (Best Poster Award).**

PROFESSIONAL & SCIENTIFIC SERVICE

- American Association of Immunologists (AAI) – Associate Member
- American Geriatrics Society (AGS) – Member
- Invited Speaker & Co-Organizer, *Systems Immunology in Aging and Complex Diseases Symposium*, The Jackson Laboratory, Farmington, CT (2024); Moderator for 2022 & 2023 sessions.

- **Peer Reviewer** for NPJ Digital Medicine, Frontiers in Immunology, Frontiers in Genetics, Frontiers in Aging, Frontiers in Endocrinology, Frontiers in Public Health, MDPI Diagnostics, and MDPI Current Oncology.
- **Guest Associate Editor,** Aging and the Immune System, Frontiers in Aging
- **Abstract Reviewer**, *Systems Immunology in Aging and Complex Diseases Symposium*, 2024.

- **Vice President (2023–2024)**, Women in Science and Engineering (WiSE) program, The Jackson Laboratory for Genomic Medicine.

TEACHING EXPERIENCE

- Teaching Assistant, "Current Trends in Drug Discovery (BC302)" – Indian Institute of Science, Bangalore, India (Jan–Mar 2016)
- Teaching Assistant, "Protein: Structure and Function (BC202)" – Indian Institute of Science, Bangalore, India (Aug–Dec 2015)
- Teaching Assistant, "Winter School on Quantitative Systems Biology" – International Centre for Theoretical Sciences, Bangalore, India (Dec 2015)